

Localized in-gap eigenstates of a 10^4 -vertex disordered photonic network from direct interior Maxwell eigensolves

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Bottom-up (611 bands, validated against MPB at 64^3) and interior (133 certified pairs near eigenvalue index 5000, validated against it) Maxwell eigensolves resolve the gap-edge modes of local-self-uniform photonic networks with 10^3 and 10^4 vertices. The 10^3 gap is empty of eigenvalues; the 10^4 pseudogap holds ten certified states, five of them bulk-localized with $\xi = 1.8\text{--}2.5\ \mu\text{m} \ll L = 24.6\ \mu\text{m}$ and persistent under periodic boundary wrapping. Decay lengths funnel to $\sim 2\ \mu\text{m}$ at the edges; level statistics are GOE-like in the bands, Poisson-compatible in 12 levels at the lower edge. No mobility edge is claimed.

Motivation.—Local-self-uniform (LSU) networks are disordered trivalent dielectric networks that open a full three-dimensional photonic band gap without long-range order [1], in the family of amorphous-diamond [2, 3] and hyperuniform [4–9] photonic materials. Gap formation and Anderson localization [10, 11] in such correlated disordered media are linked [12, 13], and a transition from diffusion to localization near the band edge has been observed in 3-D amorphous networks [14], but what the eigenmodes *inside* and at the edges of the gap look like at the 10^4 -vertex scale has not been resolved by direct eigenmode computation. Sellers *et al.* state that an N -vertex supercell has its gap between bands $N/2$ and $N/2 + 1$ [1], consistent with the band count of the parent gyroid/srs crystal [15], so at $N = 10^4$ the gap sits at eigenvalue index ≈ 5000 ; computing all lower bands bottom-up would need 1.38 TB of locked vectors at 256^3 and $O(100)$ days, hence an interior method.

System and operator.—We use the $N = 1000$ reference network of Ref. [1] (box $L = 11.44\ \mu\text{m}$, mean rod length $0.80\ \mu\text{m}$) and an $N = 10^4$ network generated with the same protocol at the same vertex density ($0.668\ \mu\text{m}^{-3}$, $L = 10^{1/3} \times 11.44 = 24.647\ \mu\text{m}$). Rods are rasterized as binary voxels (no subpixel smoothing [16, 17]) in two decorations: the “elliptical” convention of the original $N = 1000$ montage (radius $0.2252\ \mu\text{m}$ in a z -space stretched by 2.5; $n = 2.9275$, $\varepsilon = 8.57$; filling fraction $\text{ff} = 21.7\%$), and a “circular” decoration (radius $0.331836\ \mu\text{m}$, $n = 2.9$, $\varepsilon = 8.41$) whose radius was bisected to $\text{ff} = 22.00\%$ at 256^3 (22.01% at 192^3 ; 21.91% at 128^3 for $N = 1000$). Following MPB [18] we solve $\Theta \mathbf{H} = \nabla \times \varepsilon^{-1} \nabla \times \mathbf{H} = (\omega/c)^2 \mathbf{H}$ in the two-component transverse plane-wave basis of the supercell at $k = \Gamma$ (six 3-D FFTs per application; the two $\omega = 0$ modes removed exactly; Θ real symmetric on real fields). Eigenvalues are quoted as $\lambda = (\omega/c)^2$ in μm^{-2} and as $\nu = \omega a / 2\pi c$ with $a = L/5 = 2.288\ \mu\text{m}$ (density-equivalent 8-vertex cell, the same for both sizes). Band numbering is MPB’s (bands 1–2 are the $\omega = 0$ pair).

Method.—(a) For $N = 1000$ we use a deflated block LOBPCG [19, 20] with MPB’s transverse-projection

preconditioner [18], blocks of 32 with 12 guard vectors, complex64 vectors and operator on the GPU, all Gram/Rayleigh–Ritz algebra in fp64 on the host (TF32 matmuls disabled), per-band relative residual $\leq 10^{-4}$, and a host-streamed locked set. On identical 64^3 grids it reproduces MPB (fp64, tolerance 10^{-7}): $\max \Delta\omega/\omega = 9.0 \times 10^{-6}$ over 298 bands and 3.5×10^{-5} over 658 bands (medians 1.4×10^{-6}). The pre-registered convergence gate failed as registered: over $64^3/96^3/128^3$ the gap-edge band 500 is non-monotone with a 0.27% spread (three orders of magnitude above the solver–MPB error: rasterization-limited; the other five sampled bands converge monotonically) and the elliptical gap width runs $2.35 \rightarrow 1.93 \rightarrow 2.08\%$. (b) For $N = 10^4$ we locate the gap with the kernel polynomial method [21, 22] (KPM; 256^3 , degree 12 000, 12 probes, Jackson kernel, $\lambda_{\max} = 3975$) and solve the window $\lambda \in [1.757, 2.117]$ at 192^3 by two-stage bandpass Chebyshev filtered subspace iteration [23–26]: a Jackson step-difference filter of degree 3000–4000 builds a basis of $1.5\times$ the predicted count from random starts and the same filter at degree 12 000 polishes it; candidates are extracted only by fp64 Rayleigh–Ritz on the original Θ and accepted only if their relative residual on Θ is $\leq 10^{-4}$, which by Weyl’s inequality places a true eigenvalue within $10^{-4}\lambda$ of every accepted value (spurious eigenvalues are thereby excluded; eigenvectors of the two pairs closer than that are certified only as a two-dimensional span). It won a measured bake-off on a 50-band interior slice of the exact $N = 1000$ spectrum (24/50 verified within budget vs 0/50) against folded-spectrum LOBPCG [27, 28] and shift-invert PMINRES at the tested budgets; filtered Lanczos [29] was excluded on memory grounds [Supplemental Material (SM) [30]]. Its accuracy is set by re-solving that slice in the production configuration (polish degree 8000 instead of 12 000): 50/50 found, 0 ghosts, $\max \Delta\lambda/\lambda = 2.4 \times 10^{-7}$ (median 6×10^{-8})—after correcting a Rayleigh-quotient normalization bug that biased every eigenvalue high by $+4.7 \times 10^{-5}$ ($\lambda_{\text{corr}} = \lambda_{\text{raw}}/\|x\|^2$, applied retroactively; SM)—and by an independent cross-check on the circular $N = 1000$ decora-

tion (210/210 states, $\max 4.1 \times 10^{-7}$). Localization is quantified per mode from $u = \varepsilon |\mathbf{E}|^2$ by the participation fraction $p = (\sum u)^2 / (N_{\text{vox}} \sum u^2)$ and an envelope decay length ξ from a linear fit of $\ln u(r)$ vs the minimum-image distance r from the peak over $[0.10, 0.95] L/2$, $u \propto e^{-2r/\xi}$; a fit is “unresolved” (lower bound only) if $\xi \geq L/2$, if u decays by less than one decade, or if $r^2 < 0.7$.

Results, $N = 1000$.—All 611 bands (through MPB band 613) of the elliptical structure at 128^3 were solved in 5.5 h; the gap lies between bands 500 and 501 as pre-registered: $\lambda = 1.8830 \rightarrow 1.9628 \mu\text{m}^{-2}$, $\Delta\nu/\nu = 2.08\%$ at 128^3 (centre $\nu = 0.505$), a spacing $10\times$ the median of its 20 neighbours. With the circular decoration the gap is again at 500|501 and widens to $1.8276 \rightarrow 2.0225 \mu\text{m}^{-2}$, $\Delta\nu/\nu = 5.07\%$ ($26\times$ the neighbouring spacings). Both gaps are empty of eigenvalues. The regenerated montage matches the original in character (SM). Band-edge modes are localized, deep-window modes extended [Fig. 2(a)]: circular decoration $\xi = 1.47 \mu\text{m}$ (band 500, $r^2 = 0.97$, $p = 0.42\%$) and $2.30 \mu\text{m}$ (band 501, $p = 3.9\%$) vs $p = 9\text{--}37\%$ far from the gap; elliptical $\xi = 1.82$ and $2.92 \mu\text{m}$. Only 42 of 210 circular-decoration modes have resolved ξ (142 hit the $L/2 = 5.72 \mu\text{m}$ ceiling), the quantitative reason for the $N = 10^4$ computation.

Results, $N = 10^4$.—KPM places 5010.3 ± 12.5 transverse states below $\lambda = 1.957$, i.e. 0.501 bands per vertex, consistent with the $N/2$ rule to $\pm 0.3\%$ (gap at MPB bands $\approx 5012|5013 \pm 13$). The Jackson-smoothed KPM density falls below 160 states per unit λ over $[1.864, 1.996] \mu\text{m}^{-2}$ ($\Delta\nu/\nu = 3.4\%$; 5%/20% criteria: $[1.885, 1.968]/[1.837, 2.022]$, a ± 0.03 systematic), narrower than the 5.07% gap of the $N = 1000$ box with the same decoration and contained within it; the number of certified states inside the bracket is 5, 10 and 21 for the 5, 10 and 20% criteria. Three overlapping slices at 192^3 ($69 + 5 + 61$ pairs, two duplicates removed by eigenvector overlap) give **133 residual-certified eigenpairs** in $[1.75705, 2.11657] \mu\text{m}^{-2}$ (worst residual 5.9×10^{-5} ; 104 GPU-hours, 1.16×10^7 operator applications), with no in-window pair left unconverged. Completeness is certified on the sub-interval $[1.9063, 1.9606]$ by a deflated-probe KPM count of missed states, $|\text{missed}| < 0.01$ (stochastic s.e. 10^{-4} ; the estimator’s own floor from modelled leakage and eigenvector error is $\sim 3 \times 10^{-3}$, against a signal of ≈ 1 per missed state); on the full window the same estimator ($+1.7 \pm 0.3$) carries an edge-leakage bias $\geq 1.2 \pm 0.2$ and is consistent with zero or one missing state (SM). The largest interior spacing of the certified spectrum runs from 1.8860 to $1.9264 \mu\text{m}^{-2}$ ($\Delta\nu/\nu = 1.06\%$, $28\times$ the median spacing; completeness is certified only above 1.9063); but the nominal gap is not empty. **Ten certified states lie inside** $[1.864, 1.996]$, at $\lambda = 1.8690, 1.8708, 1.8731, 1.8860, 1.9264, 1.9296, 1.9441, 1.9472, 1.9738, 1.9901$ [Fig. 1(d), Table SI]. Four of them are flagged as boundary-seam states of the rasterization convention: rods whose radius pokes through

a box face are not wrapped, so the outermost voxel shell has $\text{ff} = 0.1975$ against 0.2211 inside (an 11% deficit, a planar defect), and the states at 1.8708, 1.8731, 1.9296, 1.9472 carry 18, 24, 44 and 42% of their energy in that shell (6.1% of the volume; three peak on a box edge). Re-solving the gap window with minimum-image wrapping (0.078% of voxels change) finds no partner for three of them (best overlaps 0.006, 0.131, 0.090), but the periodic solve is itself measured incomplete ($+2.0 \pm 0.4$ missed), so their absence is likely, not proven; the fourth (1.9472, overlap 0.30) is undetermined. A fifth in-gap state (1.9441) is compact ($p = 0.56\%$) but has a non-exponential envelope ($r^2 = 0.33$, ξ unresolved); its periodic partner (overlap 0.95) fits as $\xi = 2.2 \mu\text{m}$, and it is excluded from the candidate list conservatively. **Five bulk-localized in-gap states remain** among the certified set: $\xi = 1.80\text{--}2.51 \mu\text{m}$ ($\xi/L = 0.07\text{--}0.10$), $p = 0.034\text{--}0.33\%$ (participation volumes $5\text{--}49 \mu\text{m}^3$ in a $15\,000 \mu\text{m}^3$ box), $r^2 = 0.971\text{--}0.998$, shell energy $2.5\text{--}10.5\%$; all five persist under periodic wrapping with overlaps 0.968–1.000 and shifts $\Delta\lambda = -0.0007$ to -0.0030 , every shift negative as required when dielectric is added. A $2.4\times$ voxel refinement (256^3) on two anchor windows matched 11 of 11 states in $[1.84, 1.95]$ one-to-one by eigenvector overlap (min 0.873, median 0.996, the two lowest a rotating near-degenerate pair; $|\Delta\omega/\omega| \leq 0.34\%$) and 10 of 11 in $[1.99, 2.035]$ ($\leq 0.04\%$, an upper bound: 3 of 11 fine pairs certified, the rest carrying a Weyl uncertainty $\leq 0.011\%$). Three candidates (1.8690, 1.8860, 1.9264) lie in the low anchor and persist; 1.9738 was not tested, and 1.9901, at the high-anchor floor, has no fine partner. This is good evidence against a rasterization artefact for the tested states; with 7.8 voxels/ μm and the 0.27% spread seen at $N = 1000$, sub-percent gap-edge frequencies are not claimed.

The decay length varies smoothly across the window [Fig. 3(a)]: 121 of 133 fits are resolved (12 fail $r^2 \geq 0.7$, one of which also exceeds the ceiling), and the medians by spectral bin are $5.0 \rightarrow 3.0 \rightarrow 2.1 \mu\text{m}$ approaching the lower edge, $2.1 \mu\text{m}$ inside, and $2.8 \rightarrow 4.7 \rightarrow 6.1 \mu\text{m}$ climbing away above; p spans 0.03–12% [Fig. 3(b)], and the participation-volume radius $(3V_p/4\pi)^{1/3}$ follows the same funnel ($2.7 \rightarrow 1.5 \rightarrow 1.2 \mu\text{m}$ and $1.4 \rightarrow 3.1 \rightarrow 4.1 \rightarrow 5.4 \mu\text{m}$). The envelope fit is not a precision quantity: shrinking the fit range to $[0.10, 0.60] L/2$ moves ξ by a median 30% and the window-end/near-edge contrast from $1.7\text{--}2.2\times$ to $1.6\text{--}1.8\times$; the funnel is robust, three-figure ξ values are not.

Can we claim Anderson localization?—We take the items in turn. (1) *Spatial localization*: supported. The five bulk in-gap candidates have $\xi = 1.8\text{--}2.5 \mu\text{m}$ in $L = 24.6 \mu\text{m}$, $p \leq 0.33\%$ and $r^2 \geq 0.97$; the tiles [Fig. 2(b)] show single compact blobs a few μm across. ξ is an envelope-fit length with tens-of-percent fit-range sensitivity, not a transport length. (2) *Interference-driven (Anderson) vs trap-like (Lifshitz-tail [31]) states*: not sep-

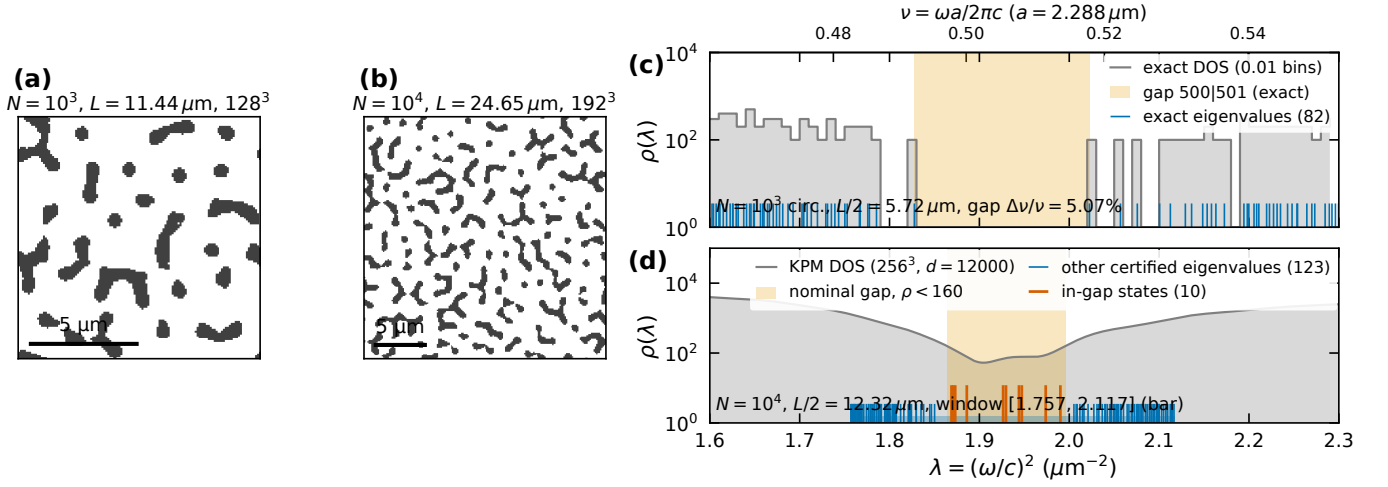


FIG. 1. (a,b) Mid-plane $\epsilon(\mathbf{r})$ of the $N = 10^3$ (128^3) and $N = 10^4$ (192^3) networks, circular decoration (dark = dielectric). (c) $N = 10^3$, circular decoration: exact eigenvalues (bottom-up, 611 bands) around the gap and their binned density of states; the shaded band is the exact gap between MPB bands 500 and 501. (d) $N = 10^4$: KPM density of states (256^3) with the 133 residual-certified 192^3 eigenvalues (blue) and the ten inside the nominal gap $\rho < 160$ states per unit λ (red); bar: solved window. Top axis: $\nu = \omega a / 2\pi c$, $a = 2.288 \mu\text{m}$.

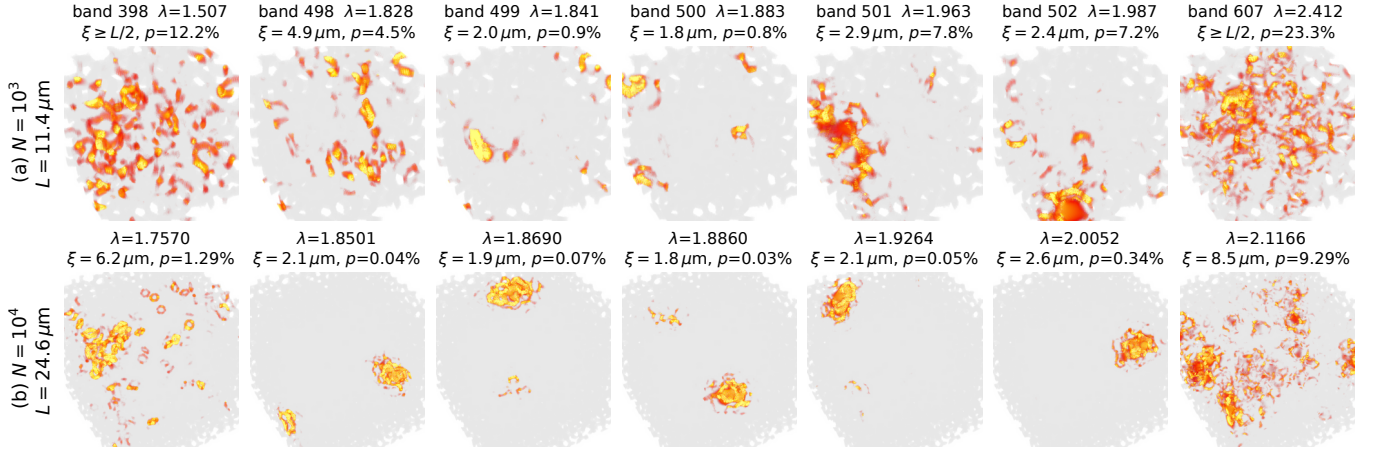


FIG. 2. Energy density $\epsilon|\mathbf{E}|^2$ (volume render, per-tile colour scale clipped at the 99.9th percentile; network in translucent grey). (a) $N = 10^3$, elliptical decoration, bands 398, 498–502, 607 around the 500|501 gap. (b) $N = 10^4$, circular decoration: window bottom (1.7570), the nearest state below the nominal gap (1.8501), three of the five in-gap candidates (1.9738 and 1.9901 not shown), the nearest state above it (2.0052), window top (2.1166). ξ and p as defined in the text; “ $\xi \geq L/2$ ” marks a ceiling-limited fit.

arable with this data. Both boxes are single realizations with the same local statistics (ff coarse-grained at $2 \mu\text{m}$: 0.2204 vs 0.2209, s.d. 0.0398 vs 0.0405), but $N = 10^4$ has ten times as many ξ -cells; at 5 in-gap states per $N = 10^4$ volume an empty $N = 1000$ gap has probability $e^{-0.5} = 0.61$ —but over the interval [1.8276, 2.0225] where the $N = 1000$ gap is actually empty the $N = 10^4$ spectrum holds 19 bulk states, giving $e^{-1.9} \approx 0.15$, so the statistics argument is plausible, not compelling. A direct test (energy-weighted local filling fraction of candidates vs bulk controls) was run and *retracted*: unmatched in mode extent, it reverses against an extent-matched null (SM). Neither origin is established. (3) *Finite-size*

scaling: two sizes only. Matched-decoration band-edge states have $\xi = 1.47$ and $2.30 \mu\text{m}$ at $L = 11.44 \mu\text{m}$ and 1.80 and $2.09 \mu\text{m}$ at $L = 24.65 \mu\text{m}$ [edge states 1.8860 and 1.9264; Fig. 4(b)], with participation volumes 6 and $58 \mu\text{m}^3$ vs 5 and $7 \mu\text{m}^3$: ξ is L -independent within the fit uncertainty, but at the estimator’s 30% systematic this two-point test has little discriminating power, the participation volumes of the upper-edge pair differ by $8\times$, two points are not a scaling analysis and 168 of 210 $N = 1000$ fits are unresolved (142 at the $L/2$ ceiling). (4) *Level statistics* [32, 33]: Θ at Γ with real ϵ is real symmetric, so the class is orthogonal: $\langle r \rangle_P = 2\ln 2 - 1 = 0.386$, $\langle r \rangle_{\text{GOE}} = 0.531$ [34, 35]. The adjacent-gap ratio (no

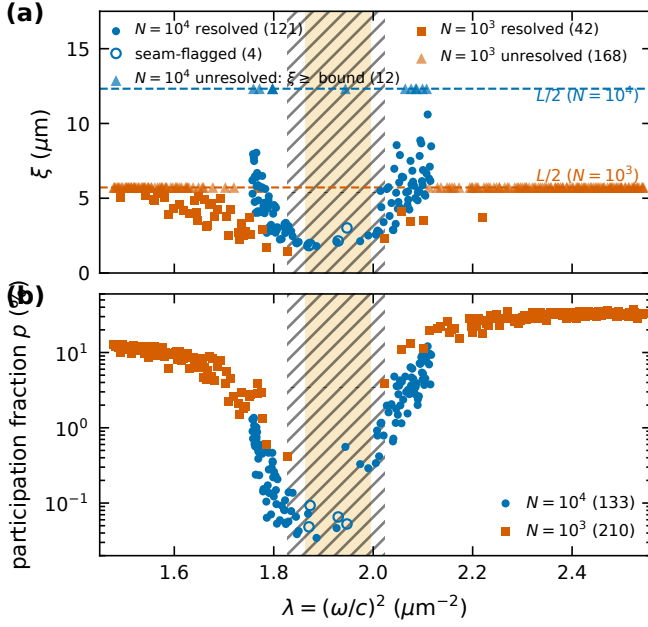


FIG. 3. (a) Envelope decay length ξ vs λ for $N = 10^4$ (circles, 192³) and $N = 10^3$ circular decoration (squares, 128³); dashed lines are the finite-size ceilings $L/2$; triangles at the ceiling are unresolved fits (lower bounds); open circles are the four seam-flagged states; shaded: nominal $N = 10^4$ gap; hatched: $N = 10^3$ gap. (b) Participation fraction. Same decoration for both sizes.

unfolding needed) is computed within contiguous bands [Fig. 4(a)]. In the extended regions it is GOE-like: 0.528 ± 0.038 (51 levels, $\lambda < 1.816$) and 0.528 ± 0.041 (42 levels, $\lambda > 2.05$) at $N = 10^4$, pooled 0.519 ± 0.024 outside the bracket; 0.508 ± 0.010 and 0.533 ± 0.011 over all 611 $N = 1000$ levels (elliptical, circular). In the 12 levels just below the gap ($1.8208 \leq \lambda \leq 1.8502$, a window chosen after inspecting the sliding-window curve), where $\xi = 2.0\text{--}3.4 \mu\text{m}$, $\langle r \rangle = 0.357 \pm 0.073$, 2.2σ below the GOE expectation for 12 levels and consistent with Poisson; four of these 12 levels carry outer-shell enhancement $> 2\times$ (up to 25%, comparable to the seam-flagged in-gap states) and are not excluded. Above the gap the 18 nearest levels ($2.005\text{--}2.050$, $p \approx 1\%$, ξ median $3.5 \mu\text{m}$) give 0.559 ± 0.047 : comparably compact states show repulsion there, an asymmetry the data do not explain. The ten in-gap levels give 0.31 ± 0.08 , but ten levels, four of them seam-flagged, are not a statistic. Uncertainties are bootstrap. The residual gate guarantees $|\Delta\lambda| \leq 10^{-4} \lambda \approx 1.9 \times 10^{-4} \mu\text{m}^{-2}$ per level (Weyl), 14% of the median spacing $1.4 \times 10^{-3} \mu\text{m}^{-2}$, while the accuracy measured on the $N = 1000$ ground truth is $\Delta\lambda/\lambda \leq 4 \times 10^{-7}$ and duplicate states found by two slices agree to 10^{-8} ; a single missing level would move the 12-level ratio by $\lesssim 0.03$. Level repulsion where states are extended and its loss where they are localized is the expected pattern, but the near-edge samples

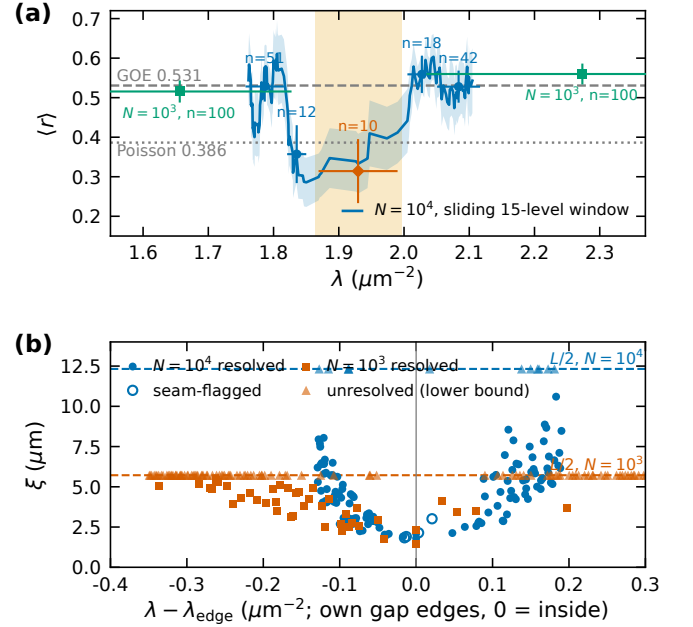


FIG. 4. (a) Adjacent-gap ratio $\langle r \rangle$: $N = 10^4$ sliding 15-level window (line, ± 1 s.e.; inside the bracket it includes the seam-flagged levels and ratios dominated by the two largest spacings, and is not interpreted) and contiguous bands (circles: 51, 12, 18, 42 levels; diamond: ten in-gap levels; bootstrap errors); $N = 10^3$ circular, bands 401–500 and 501–600 (squares). Dashed/dotted: GOE 0.531 / Poisson 0.386. (b) Matched-decoration comparison: resolved ξ vs distance from each box's own gap edges ($N = 10^3$: 1.8276|2.0225; $N = 10^4$: 1.8860|1.9264; in-gap states at 0).

are small, post hoc and seam-contaminated: suggestive, not conclusive. A Thouless-type sensitivity to boundary twists cannot be obtained from the saved Γ -point data. (5) *Mobility edge*: not claimable—no transport, one realization per size, no scaling; the $\xi(\omega)$ funnel (Fig. 3) and the $\langle r \rangle$ drop [Fig. 4(a)] are the *phenomenology* of a localization–delocalization crossover at the gap edge, the regime probed experimentally in 3-D amorphous networks [14] and predicted for correlated disorder [13]. (6) *Vector-wave caveat*: early reports of 3-D localization of light [36] were not confirmed, vector waves do not localize in uncorrelated point-scatterer ensembles [37–39], and numerical 3-D localization has been shown only for overlapping metallic, not dielectric, spheres [40]; localization at the edge of a photonic gap in a correlated network [12, 13, 41] is a different mechanism; this data addresses only the spatial structure of the eigenstates at such an edge.

Verdict.—The defensible claim is: strongly spatially localized in-gap and gap-edge eigenstates with $\xi \ll L$, with the two resolved $N = 1000$ band-edge values in the same range, consistent with localization at the edges of a photonic pseudogap in a correlated disordered network, accompanied by GOE-like level statistics in the extended

bands and a Poisson-compatible ratio in 12 post-hoc levels at the lower edge that is not reproduced above the gap. Whether these states are interference-driven (Anderson-type) or trapped in rare local configurations (Lifshitz-tail-type) cannot be decided from this data. The claim of an Anderson transition or a mobility edge requires finite-size scaling over three or more sizes and/or transport or larger-sample level statistics that this data does not contain.

Limitations.—Gap-edge frequencies are rasterization-limited (0.27% non-monotone spread at $N = 1000$; $\leq 0.34\%$ between 192^3 and 256^3 at $N = 10^4$), and on MPB’s interpolated read-back of the 64^3 grid the elliptical 500|501 spacing is 0.60% (largest at 501|502), so the gap position at 64^3 is convention-dependent. Decay lengths are bounded by $L/2$ and fit-range dependent at the 30% level; one disorder realization per size. Arithmetic is fp32 on the device with fp64 Rayleigh–Ritz; the merged 133-mode set is orthonormal only to 4.9×10^{-4} across independently solved slices (8.6×10^{-6} within), the level the 10^{-4} residual gate permits, which affects no spectrum, montage or localization result. Twenty of the 133 window modes have outer-shell enhancement $> 2\times$; only the four inside the bracket were re-solved seam-free, the others enter every ξ , p and $\langle r \rangle$ result. Of the statements withdrawn during the project (eight are listed in the SM), four bear on the results here and are absent: the rare-region z -test, the statement that the resolution confound is “closed” (its power-retention statistic is saturated), the “vanished” verdict for the state at 1.9472, and all uncorrected eigenvalues. Full records, the gate ledger and every number with its source file are in the SM [30].

[Acknowledgments to be filled.]

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 - [30] See Supplemental Material at [URL will be inserted by publisher] for the complete technical record: structure generation and rasterization (including the boundary-seam artefact), operator and solver details, all pre-registered gates with outcomes, the bake-off, the Rayleigh-normalization correction, completeness checks, localization methodology, level-statistics details, the full 133-state table, retracted claims and the full list of limitations.
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